

Review retrospective:

$\kappa_3 = 0$ for affine observables on the harmonic Gibbs — a clean Gaussian-cumulant pass with one dead open

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Abstract

A soundness review of `Laplace/OneD/HarmonicAffineKappa3.lean`: the theorem that the third cumulant of any triple of *affine* observables vanishes under the 1D harmonic Gibbs measure — the multilinearity strict-improvement of the identity-observable case. Result: **a clean fidelity pass with a single edit**, the removal of a dead open `Real`. The independent reviewer re-derived every moment value and polynomial coefficient and found no discrepancy.

1 Setting

This continues the harmonic 1D stack reviewed across the same loop — the regularity witness (`HarmonicGibbsRegularity`) and the monomial observable witnesses (`HarmonicGibbsObservableMonomials`). The file under review proves a cumulant identity rather than a regularity datum: for the harmonic potential $L(x) = (\lambda/2)x^2$ with $\lambda, t > 0$, the third cumulant

$$\kappa_3(b_1x + c_1, b_2x + c_2, b_3x + c_3) = 0$$

for arbitrary affine coefficients. It is the multilinearity lift of `Threepoint's kappa3_harmonic_id_id_id.eq.zero` (the (x, x, x) case).

2 What was reviewed

The headline `kappa3_harmonic_affine_eq.zero`; the private bridge `threepoint_gibbsExp_volume_zero.eq` ($h = 0$ unperturbed `Threepoint.gibbsExp` → `Laplace.gibbsExpectation`); the harmonic moment wrappers ($m_1 = 0$, $m_2 = 1/(\lambda t)$, $m_3 = 0$); the integrability helpers; and the degree- ≤ 3 linearity lemma `gibbsExp_poly_le3_harmonic` giving $\langle a_3x^3 + a_2x^2 + a_1x + a_0 \rangle = a_2/(\lambda t) + a_0$. The informal corpus was the file's docstrings, the `Threepoint.kappa3` definition ($\kappa_3 = \langle \varphi AB \rangle - \langle \varphi A \rangle \langle B \rangle - \langle \varphi B \rangle \langle A \rangle - \langle AB \rangle \langle \varphi \rangle + 2\langle \varphi \rangle \langle A \rangle \langle B \rangle$, at $h = 0$), and the identity-case headline it improves.

3 Fidelity / soundness findings

Sound; no defect. The mathematics is the Gaussian cumulant fact: the harmonic Gibbs weight $e^{-t(\lambda/2)x^2} = e^{-(\lambda t/2)x^2}$ is a centred Gaussian of variance $1/(\lambda t)$, whose cumulants of order ≥ 3 all vanish; κ_3 is multilinear in (φ, A, B) and invariant under adding constants, so affine observables

give 0. The Lean proof realises this concretely: unfold the five-term κ_3 through the $h = 0$ bridge, expand each affine product into a degree-3 polynomial, substitute $m_1 = m_3 = 0$ and $m_2 = 1/(\lambda t)$, and `ring`. The reviewer independently re-derived and confirmed: the moment formula $a_2/(\lambda t) + a_0$; each moment value; *every* affine-product coefficient (the cubic $b_1 b_2 b_3$, the quadratic $b_1 b_2 c_3 + b_1 c_2 b_3 + c_1 b_2 b_3$, the linear $b_1 c_2 c_3 + c_1 b_2 c_3 + c_1 c_2 b_3$, and the three pairwise expansions); and that the call’s slot mapping $(\varphi, A, B) = (b_1 x + c_1, b_2 x + c_2, b_3 x + c_3)$ matches the definition. “No fidelity discrepancies found.”

4 Simplifications applied

One dead open. `open Real` carried nothing: every `Real`-namespace identifier is qualified (`Real.exp` throughout), and the bare `exp/√/π` tokens a `grep` turns up live only inside comments and docstrings. `open MeasureTheory` stays (bare `volume`, `Integrable`, `Measure`). Build green, no in-file warnings, public signatures byte-identical; the reviewer affirmed. All five imports are load-bearing and were kept — notably `integrable_pow_mul_exp_neg_mul_sq` genuinely lives in the seabed’s `IntegralRemainder.lean` (the *pow* variant, not Mathlib’s `rpow` one), so unlike the previous review’s import this one is not dead.

5 Disagreements and open questions

None. The reviewer offered a sheaf of proof-golf and refactor ideas — one-line `simp` rewrites of the moment and integrability wrappers, inlining `harmonic_Z_ne_zero`, dropping `gibbsExp_affine_harmonic` as a special case of the degree-3 lemma, and the `show...funext;ring` verbosity — all declined as either internal-shape-only (and `simp`-set golf has a high break rate) or invasive structural refactors that erase proof-documenting structure. One genuine name-clarity nit is recorded report-only: the private `gibbsExp.*` lemmas state facts about `Laplace.gibbsExpectation`, not `Threepoint.gibbsExp`, so a `laplaceGibbsExp.*` prefix would read truer — but a multi-call-site rename is a nicety, not a defect (the same call the AnharmonicKappa3 review made on its `*_affine_*→*_linear_*` rename).

6 Lessons

- **The `pow` vs `rpow` integrability lemmas have different homes, and that flips the dead-import verdict.** The immediately-preceding review found `integrable_rpow_mul_exp_neg_mul_sq` to be Mathlib’s (making its `IntegralRemainder` import dead); here `integrable_pow_mul_exp_neg_mul_sq` is the seabed’s own lemma in `IntegralRemainder.lean`, so the same-named import is load-bearing. The one-letter difference in the lemma name is the whole story — `grep` the *definition site*, do not pattern-match on the import line.
- **A whole-Gaussian fidelity check reduces to checking the moments and the coefficients.** For a cumulant-vanishing theorem the only places a real error could hide are a wrong moment value ($m_2 \neq 1/(\lambda t)$, or a non-zero odd moment) or a slipped polynomial coefficient; the reviewer’s value-add was re-deriving exactly those, not the (manifestly correct) Gaussian story.

7 Follow-ups

- The remaining never-reviewed laplace OneD files: `HarmonicPerturbedMoments`, `QuarticBoundedPrior`, `QuarticBoundedPriorTestFn`, and the large `IntegralRemainder`.
- Optional (report-only): the `gibbsExp_*`→`laplaceGibbsExp_*` private rename.